

Paper Manager pack — OpenVLA (Oct 21)

Paper: OpenVLA: An Open-Source Vision-Language-Action Model

arXiv: <https://arxiv.org/abs/2406.09246> · CoRL 2024

HF: [openvla/openvla-7b](#) · <https://openvla.github.io>

PDF: </workspace/papers/openvla-2406.09246.pdf>

Session: Oct 21 · due 9:30 AM · **You = Empiricist (seat 1)** · Archaeologist=5

Sources: Summarizer + Senior Research (</workspace/openvla-quiet/DELIVERABLE.md>)

1. Paper summary

Open Llama/Prismatic **discrete-AR VLA** on OXE ~**970k** with **256-bin** EE actions, **no chunking**. Prismatic (fused **DINOv2+SigLIP + Llama-2 7B**). Beats RT-2-X on WidowX/Google at **7× smaller**; LoRA/int4 make consumer FT/serve practical.

Syllabus role: historical open AR baseline — BEHAVIOR deploy belongs to $\pi_{0.5}$ /OpenPI/GR00T.

2. Action-family taxonomy (CLEAR)

Family	Model	Signature
Naïve discrete AR bins	**OpenVLA / RT-2**	Lang grounding @ ~5–6 Hz; weak high-Hz
DDPM chunks	**Octo / DP**	Smoother narrow skills
FM chunks	** π_0 / $\pi_{0.5}$ **	2025–26 deploy winners
Compressed discrete AR	**FAST / OpenVLA+FAST / π_0 -FAST**	AR recovers high-Hz

3. Keywords / stack

DINOv2||SigLIP + Llama-2 7B @ 224²; vision unfrozen; 7D EE; 256 bins; CE; single RGB; **no** wrist/proprio/history/chunks. Train ~21.5k A100-h. Infer ~6 Hz / 4090.

4. Role packs

Stakeholder (18+3)

Open HF 7B for lang manip + LoRA on 1×A100 / int4 ~7GB. 2026: teaching + ablation harness, not BEHAVIOR entrant — ship $\pi_{0.5}$ /GR00T.

Scientific Reviewer (8+3)

Strengths: open 7B+OXE; PEFT/quant tables; honest limits. Weaknesses: RT-2-X comparison quirks; DP still better narrow dexterity; LIBERO deltas small. Verdict: strong CoRL’24 open-science systems paper — not 2026 substrate.

Empiricist (11+3) — **YOU (seat 1)** — HIGH PRIORITY

Priority	Experiment
P0 Y	HF ckpt dry infer (~15–17 GB bf16); int4/int8 serve (~7 GB int4)
P0 Partial	LIBERO-Spatial LoRA r=32 scout
P1	LoRA r sweep; full FT vs LoRA; vision FT vs freeze; **AR discrete vs DP**; OpenVLA vs π_0 FM vs Octo @ matched commit_s
P2	no-op filter; FAST drop-in vs 256 bins
Defer	21.5k A100-h OpenX re-pretrain; multi-robot real suites

H: LoRA r=32 $\approx$ full FT within ~2 pp; unfreeze vision helps; OpenVLA AR wins multi-object lang; DP/FM win precision/smooth Hz.

Tattoo: WidowX **70.6%** vs RT-2-X 50.6 / Octo 20; Google **85.0%**; LoRA **68.2** $\approx$ full **69.7**; LIBERO FT **76.5**.

Ladder: HF smoke → LoRA r=32 → DP-matched side-by-side → optional FAST+ before high-Hz claims.

Env: /workspace/opencvla-quiet/env_outline/.

Archaeologist (11+3) — weight 5

Priors: RT-1/RT-2; Prismatic; Llama-2; SigLIP+DINOv2; OXE; Octo peer; DP foil.

Genealogy: RT-2-style discrete AR → **OpenVLA** (open) || Octo (diffusion) → π_0 (FM) → FAST upgrades naïve bins → $\pi_{0.5}$ /OpenPI/BEHAVIOR.

Visionary (11+3)

5. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Open HF + LoRA, not 2026 entrant
Reviewer	8+3	Open-science strengths + comparison caveats
Empiricist	**11+3**	**LoRA ladder + AR vs FM/DP (HIGH)**
Archaeologist	11+3	Discrete AR peer era + FAST upgrade
Visionary	11+3	Harness vs deploy split

Spine: action-family taxonomy; no-chunk AR latency. PDF for Mac.

GitHub title: [ECE 605] - OpenVLA Making Empiricist

Paths

Artifact	Path
This pack	`/workspace/opencvla-quiet/KANTA_PACK.md`
Senior Research	`/workspace/opencvla-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/opencvla-2406.09246-summary.md`
Paper PDF	`/workspace/papers/opencvla-2406.09246.pdf`